



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

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Experiment-3

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Subject Name: ADBMS

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Section/Group: KRG-2B

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1. Aim:

Department Salary Champions Explorer

In a bustling corporate organization, each department strives to retain the most talented (and well-compensated) employees. You have access to two key records: **one lists every employee along with their salary and department, while the other details the names of each department.** Your task is to identify the **top earners in every department.**

If multiple employees share the same highest salary within a department, all of them should be celebrated equally. The final result should present the **department name, employee name, and salary of these top-tier professionals** arranged by department.

Merging Employee Histories: Who Earned Least? (Hard)

Two legacy HR systems (A and B) have separate records of employee salaries. These records may overlap. Management wants to **merge these datasets** and identify **each unique employee** (by EmpID) along with their **lowest recorded salary** across both systems.

Objective

1. Combine two tables A and B.
2. Return each EmpID with their **lowest salary**, and the corresponding **Ename**.



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2. Objective:

- To understand and implement sub-queries in SQL.
- To identify the top earners in each department using correlated sub-queries.
- To practice handling scenarios where multiple employees share the same maximum salary.
- To merge datasets from multiple sources using `UNION ALL`.
- To apply `GROUP BY` with aggregate functions (`MAX`, `MIN`) for meaningful reporting.
- To retrieve the lowest recorded salary for each employee across different systems.
- To develop practical problem-solving skills for analytical database queries.

3. DBMS Script :

```
USE KRG_2B;
```

```
--EXPERIMENT 03: Department Salary Champions Explorer (MEDIUM LEVEL)
```

```
CREATE TABLE department (
```

```
    id INT PRIMARY KEY,
```

```
    dept_name VARCHAR(50)
```

```
);
```

```
CREATE TABLE employee (
```

```
    id INT,
```

```
    name VARCHAR(50),
```

```
    salary INT,
```

```
    department_id INT,
```

```
    FOREIGN KEY (department_id) REFERENCES department(id)
```

```
);
```

```
INSERT INTO department (id, dept_name) VALUES
```

```
(1, 'IT'),
```

```
(2, 'SALES');
```

```
INSERT INTO employee (id, name, salary, department_id) VALUES
```



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```
(1, 'JOE', 70000, 1),  
(2, 'JIM', 90000, 1),  
(3, 'HENRY', 80000, 2),  
(4, 'SAM', 60000, 2),  
(5, 'MAX', 90000, 1);
```

```
SELECT (SELECT dept_name FROM department d where d.id = e.department_id) AS  
DEPT_NAME, name, salary  
FROM Employee e  
WHERE salary IN (SELECT MAX(e2.salary) FROM employee e2 WHERE e2.department_id =  
e.department_id);
```

--EXPERIMENT 03: Merging Employee Histories: Who Earned Least? (Hard)

```
CREATE TABLE A( empid integer, Ename VARCHAR(20), Salary INTEGER);  
CREATE TABLE B(empid integer, Ename VARCHAR(20), Salary INTEGER);
```

```
INSERT INTO A VALUES  
(1,'AA',1000),  
(2,'BB',300);
```

```
INSERT INTO b VALUES  
(2,'BB',400),  
(3,'CC',100);
```

```
SELECT EMPID,Max(ENAME) AS ENAME,MIN(SALARY) AS SALARY  
FROM(  
SELECT * FROM A  
UNION ALL  
SELECT * FROM B
```

) AS INTERMEDIATE_RESULT

GROUP BY empid;

4. Output:

Output 1:

	DEPT_NAME	name	salary
1	SALES	HENRY	80000
2	IT	MAX	90000
3	IT	JIM	90000

Output 2:

	EMPID	ENAME	ESALARY
1	1	AA	1000
2	2	BB	300
3	3	CC	100

5. Learning Outcomes:

- Successfully implemented sub-queries to extract top salary earners by department.
- Practiced combining two datasets with UNION ALL.
- Used GROUP BY and aggregate functions (MAX, MIN) to derive meaningful insights.
- Understood how to merge historical records and identify minimum salaries.
- Strengthened SQL querying skills for analytical use cases.